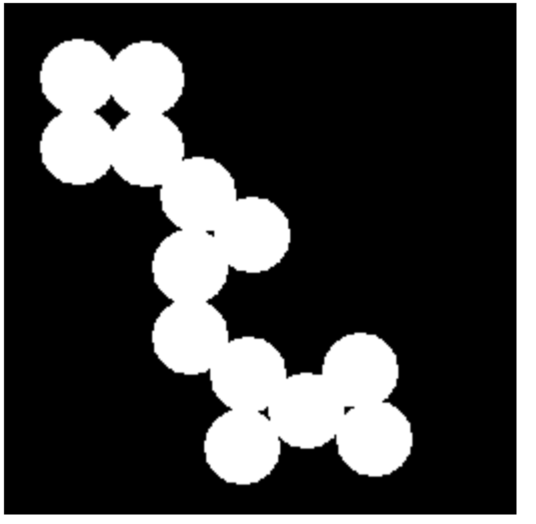
Binary Image Processing

Binary Images

- "Binary" means
  - 0 or 1 values only
  - Also called "logical" type (true/false)



- Obtained from
  - Thresholding gray level images
  - Or, the result of feature detectors

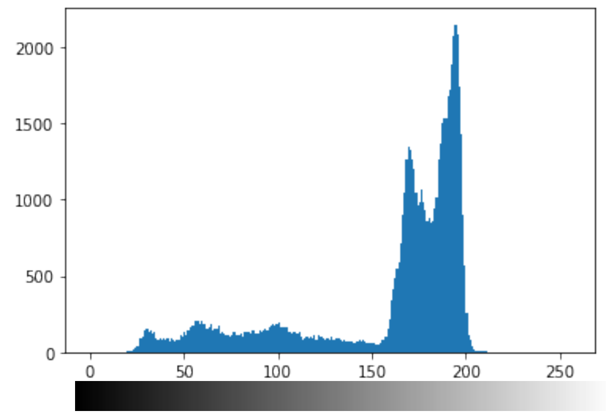
- Often want to count or measure shape of 2D binary image regions

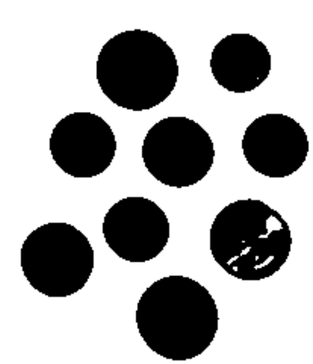
Thresholding

- Convert gray scale image to binary (0s and 1s)
- Simplifies processing and computation of features
- Can use a single threshold value (global) or a local value (adaptive)

Thresholding in OpenCV:

```
(T, threshImg) = cv2.threshold(blurred,
155, 255, cv2.THRESH_BINARY)
```





Otsu's Method for Global Thresholding

Used in Matlab's "graythresh" function

- Choose threshold to minimize the variance within groups

$$\sigma_W^2 = P_1\,\sigma_1^2 + P_2\,\sigma_2^2$$

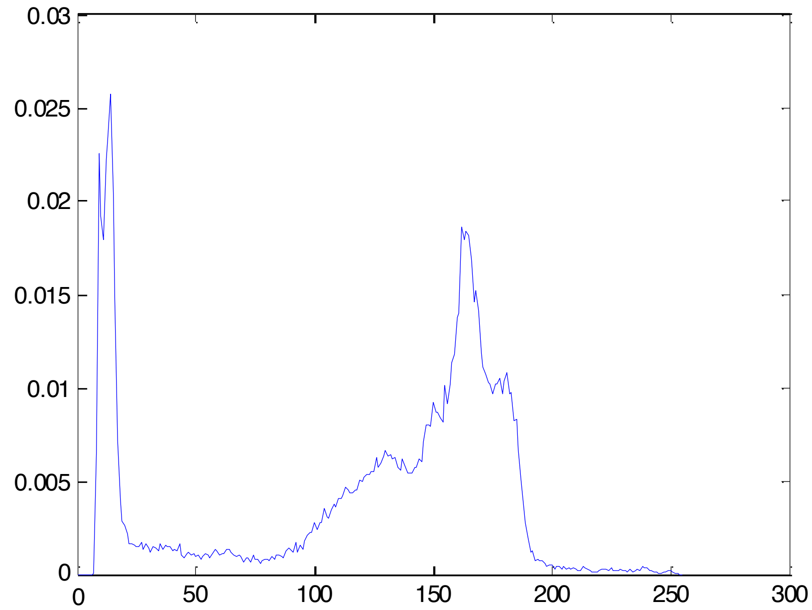
- Or equivalently, maximize the variance between groups

$$\sigma_B^2 = P_1\left(m_1 - m_G\right)^2 + P_2\left(m_2 - m_G\right)^2$$

- where

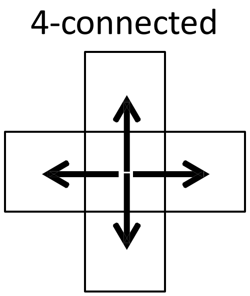
$$P_1 = \sum_{i=0}^{k} p_i, \quad P_2 = \sum_{i=k+1}^{L-1} p_i$$

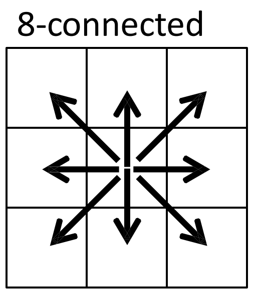
- m_G is the global mean; m_k is the mean of class k



Connected Components

- Define adjacency
  - 4-adjacent
  - 8-adjacent





- Two pixels are connected in S if there is a path between them consisting entirely of pixels in S

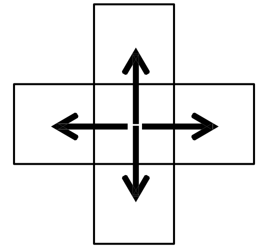
- S is a (4- or 8-) connected component (“blob”) if there exists a path between every pair of pixels

- “Labeling” is the process of assigning the same label number to every pixel in a connected component

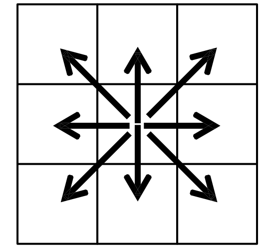
Example

- Hand label simple binary image

4-connected



8-connected



	1	1				
	1	1				
			1	1		
	1		1	1		
	1		1			1

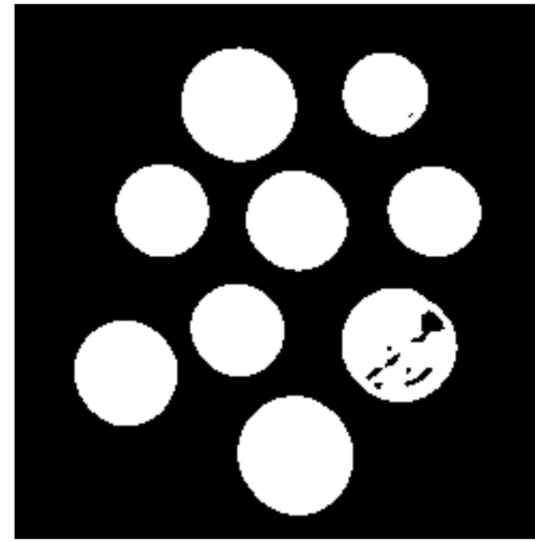
Binary image

Labeled image (4-connected)

Labeled image (8-connected)

OpenCV Example

- Labeling connected components
  `(numLabels, labels, stats, centroids) = cv2.connectedComponentsWithStats`
  - The total number of unique labels (i.e., number of total components) that were detected
  - A mask named labels has the same spatial dimensions as our input thresh image. For each location in labels, we have an integer ID value that corresponds to the connected component where the pixel belongs.
  - stats: Statistics on each connected component, including the bounding box coordinates and area (in pixels).
  - The centroids (i.e., center) (x, y)-coordinates of each connected component.



How many coins are there?

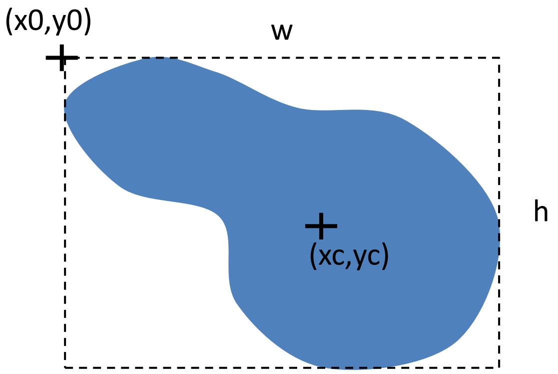
OpenCV Example

- Labeling connected components

```
(numLabels, labels, stats, centroids) =
cv2.connectedComponentsWithStats
```

  - stats: Statistics on each connected component, including the bounding box coordinates and area (in pixels).
  - The centroids (i.e., center) (x, y)-coordinates of each connected component.

- Centroid is represented as [xc, yc]
- Bounding box is represented as [x0 y0 w h], where
  - x0,y0 are the coordinates of the upper left point
  - w,h are the width and height



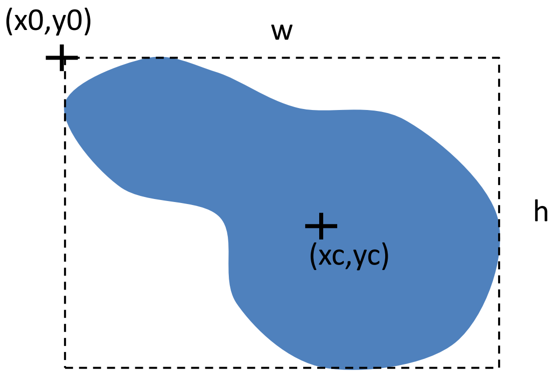
OpenCV Example

- Labeling connected components

```
(numLabels, labels, stats, centroids) =
cv2.connectedComponentsWithStats
```

  - stats: Statistics on each connected component, including the bounding box coordinates and area (in pixels).
  - The centroids (i.e., center) (x, y)-coordinates of each connected component.

- Basic features
  - Area $$A = \sum_{(r,c)\in R} 1$$
  - Centroid $$\bar{r} = \frac{1}{A}\sum_{(r,c)\in R} r, \quad \bar{c} = \frac{1}{A}\sum_{(r,c)\in R} c$$



Binary Image Morphology

- Operations on binary images:
  - dilation and erosion
  - opening and closing
- Can be used to “clean up” image
  - shrink and expand regions
  - eliminate small regions or holes
- Operations are performed with a “structuring element” S
  - a small binary image
  - like a filter mask

Dilation

1	1	1
1	**1**	1
1	1	1

S

- Defined as

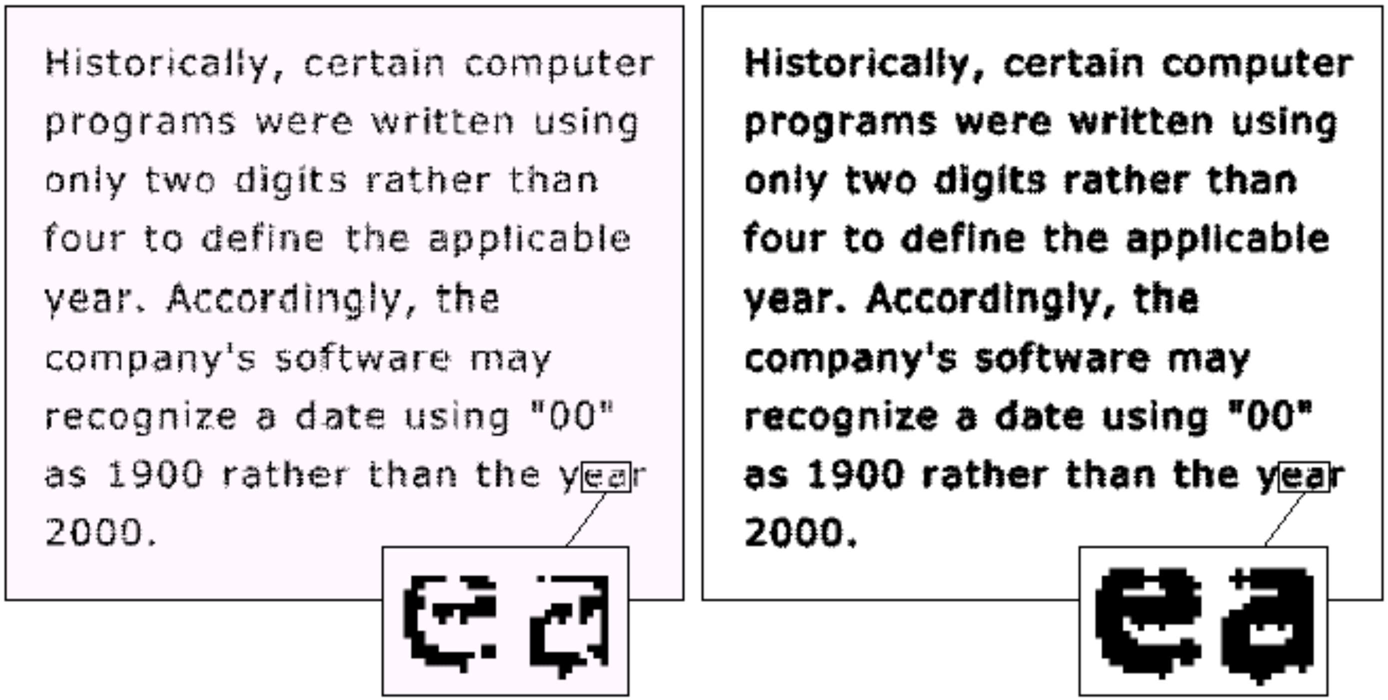
$$B \oplus S = \bigcup_{b \in B} S_b$$

- where
  - S_b is the structuring element S, shifted to b
- Procedure
  - Sweep S over B
  - Everywhere the origin of S touches a 1, OR S with B
- Expands regions

		1		1	
		1	1		

B

$B \oplus S$



0	1	0
1	1	1
0	1	0

a c
b

FIGURE 9.5
(a) Sample text of poor resolution with broken characters (magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

Erosion

1	1	1
1	1	1
1	1	1

S

- Defined as

$$B \Theta S = \{ b \mid b + s \in B, \forall s \in S \}$$

- Procedure
  - Sweep S over B
  - Everywhere S is completely contained in B, output a 1 at the origin of S
- Shrinks regions

	1	1	1	1	1
	1	1	1	1	1
	1	1	1	1	1
	1	1	1	1	

B

$B \Theta S$

OpenCV

- To create a structuring element

```
# Rectangular Kernel
>>>cv.getStructuringElement(cv.MORPH_RECT,(5,5))
array([[1, 1, 1, 1, 1],
[1, 1, 1, 1, 1],
[1, 1, 1, 1, 1],
[1, 1, 1, 1, 1],
[1, 1, 1, 1, 1]], dtype=uint8)
```

- Dilation

```
dilation = cv.dilate(img,kernel,iterations = 1)
```

- Erosion

```
erosion = cv.erode(img,kernel,iterations = 1)
```

Openings and Closings

- Opening
  - Erosion followed by dilation
  - Eliminate small regions and projections

$$B \circ S = (B \Theta S) \oplus S$$

- Closing
  - Dilation followed by erosion
  - Fill in small holes and gaps

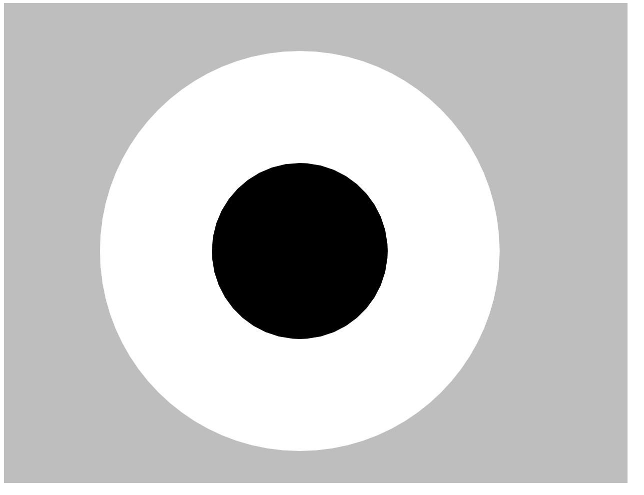
$$B \bullet S = (B \oplus S) \Theta S$$

- OpenCV functions:
  - `opening = cv.morphologyEx(img, cv.MORPH_OPEN, kernel)`
  - `closing = cv.morphologyEx(img, cv.MORPH_CLOSE, kernel)`

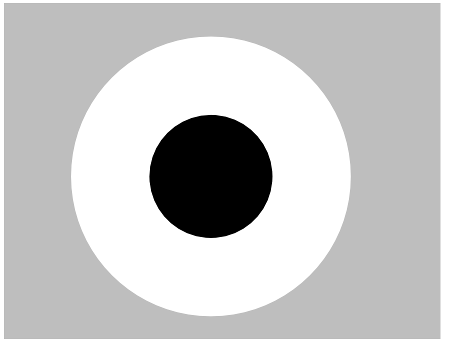
Concentric contrasting circle (CCC) target

- The target is a white ring surrounding a black dot
- This feature is fairly unique in the image, because the centroid of the white ring will coincide with the centroid of the black dot
- You can automatically find the target by finding a white region whose centroid coincides with the centroid of a black region

image "robot.jpg" on course website



CCC targets (continued)



- For more discrimination power, you can also place constraints on the binary regions, e.g.,
  - The white area must be > the black area)
  - The white bounding box must enclose the black bounding box
- Sometimes local thresholding is better than global thresholding; especially when the lighting varies across the image
- For these simple black-and-white targets, a easy way to do local thresholding is:
  - Apply an averaging filter to the image (whose size is greater than the size of the target). This creates an image of local means.
  - Subtract the local average image from the original.
  - Threshold at zero.

Example

- See if your program can locate the CCC target in each image of the video “oneCCC.mp4” on the course website